



DEPARTMENT OF MATHEMATICS

Sub Code:	CS/IS/AI/AD/CI/CY31	Sub:	Laplace Transforms and Vector Space	Test:	Test - II
Time:	03.00pm to 04.00pm	Term:	15-04-2024 TO 27-07-2024	Marks:	30
Date:	20.01.2025	Semester:	IV	Section:	CS,IS,AI,ML, AIDS, CS(AI,ML), CS(CS)

Note: Answer any TWO full questions. Each main question carries 15 marks.

Q.No.	Questions	Blooms Level	CO's	Marks
1.	(a) Determine whether the set $\{(1, 1, 1), (0, 1, 2), (3, 0, 1)\}$ forms a basis for $\mathbb{R}^3$ or not?	L1	CO3	5
	(b) Determine the matrix that defines a reflection in the X-axis followed by rotation through an angle $\pi/2$ followed by a contraction of factor $1/2$. Find the image of the point $\begin{bmatrix} 4 \\ 1 \end{bmatrix}$ under this transformation.	L2	CO3	5
	(c) Verify Rank-Nullity theorem for the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by $T(x, y) = (3x, x - y, y)$.	L4	CO3	5
2.	(a) Find the complete solution of $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.	L1	CO4	5
	(b) Let $y = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$, $u = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$. Find the orthogonal projection of y onto u . Then write y as the sum of two orthogonal vectors, one in $\text{span}\{u\}$ and one orthogonal to u .	L2	CO4	5
	(c) Find QR factorization of $A = \begin{bmatrix} -2 & 3 \\ 5 & 7 \\ 2 & -2 \\ 4 & 6 \end{bmatrix}$.	L3	CO4	5
3.	(a) Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$.	L1	CO5	5
	(b) Prove that the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $T(x, y) = (2x, x - y)$ is linear. Find the images of the elements $(1, 2)$ and $(-1, 4)$ under this transformation.	L2	CO3	5
	(c) Find the least-squares solution of $Ax = b$ where $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$.	L3	CO4	5